Detecting Hallucinations in Medical Question Answering: A Comparative Study of BERT and BioBERT

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Abstract

Hallucinations in large language models (LLMs) pose serious risks in medical applications. This work studies hallucination detection as a binary classification problem over question–answer pairs. Using the MedHallu dataset for training and Med-HALT for out-of-domain evaluation, this project compares a general-purpose BERT model with a domain-specific BioBERT model. While both models achieve strong in-domain performance, BioBERT significantly improves generalization, increasing F1 from 0.33 to 0.59 on Med-HALT. However, both models fail on adversarial "fake question" tasks, highlighting persistent limitations in robustness. These results demonstrate that domain-specific pretraining improves transfer but does not fully resolve reasoning gaps.

1 Introduction

Large language models frequently generate hallucinated content—fluent but factually incorrect outputs—which is especially problematic in medical contexts. Detecting such hallucinations is essential for safe deployment.

This project formulates hallucination detection as a supervised classification task: given a question, context, and answer, predict whether the answer is hallucinated. The model is trained on MedHallu [1], a dataset with fine-grained hallucination annotations, and evaluated on Med-HALT [2], which introduces diverse reasoning-based tasks.

A key objective of this work is to determine whether domain-specific pretraining improves hallucination detection, particularly under distribution shift. To address this, BERT and BioBERT [3] are compared under identical training conditions.

2 Related Work

Recent work has highlighted hallucination as a central limitation of LLMs. The MedHallu dataset [1] introduces structured hallucination annotations for medical QA, categorizing errors such as misinterpretation and incomplete reasoning. MedHALT [2] proposes stress tests that evaluate model reliability under challenging scenarios including misleading answers, "none of the above" reasoning, and nonsensical questions.
SelfCheckGPT [5] explores hallucination detection without supervision by measuring self-consistency, while domain-specific models such as BioBERT [3] and ClinicalBERT [4] improve performance in biomedical NLP tasks through specialized pretraining.

3 Methodology

3.1 Task Formulation

Hallucination detection is modeled as a binary classification task. Each example consists of a question (with optional context) and a candidate answer. The model predicts whether the answer is grounded (label 0) or hallucinated (label 1).

3.2 Datasets

MedHallu [1] The MedHallu dataset is used for training and in-domain evaluation. The artificial split (9,000 samples) is used for training, while the labeled split (1,000 samples) is divided equally into validation and test sets.
Each example produces two training instances (ground truth and hallucinated answer), resulting in:

- Train: 18,000 examples
- Validation: 1,000 examples
- Test: 1,000 examples

The test distribution is as follows:

- Misinterpretation of Question: 742
- Incomplete Information: 222
- Mechanism/Pathway Misattribution: 34
- Methodological Fabrication: 2

Med-HALT [2] Med-HALT is used for out-of-domain evaluation and includes reasoning-based tasks:

- False Confidence Test (FCT): 37,564 samples
- None of the Above Test (Nota): 75,464 samples
- Fake Questions Test (FQT): 1,858 samples

The total dataset size is 114,886 examples.

3.3 Models

Two models are evaluated:

- BERT-base-uncased
- BioBERT-base-cased (biomedical pretraining) [3]

Both models are fine-tuned as sequence pair classifiers.

3.4 Training Setup

Input sequences are constructed as:

Question + Context paired with Answer

Training is performed for 5 epochs with a batch size of 8 and maximum sequence length of 256. Performance is evaluated using accuracy, precision, recall, and F1 score.

4 Results

4.1 In-Domain Performance (MedHallu)

The BERT model achieves strong performance on MedHallu:

- Accuracy: 0.949
- F1 Score: 0.949

BioBERT improves further:

- Accuracy: 0.957
- F1 Score: 0.956

Per-category results show consistently high performance across hallucination types, with near-perfect scores on smaller categories such as mechanism misattribution.

Category	BERT F1	BioBERT F1
Misinterpretation	0.907	0.956
Incomplete Info	0.902	0.949
Mechanism	0.911	1.000
Methodological	0.930	1.000

Table 1: Per-category performance on MedHallu

4.2 Out-of-Domain Performance (Med-HALT)

Performance drops significantly when evaluated on Med-HALT.
BERT results:

- Accuracy: 0.432
- F1 Score: 0.327
- Recall: 0.206

BioBERT shows notable improvements:

- Accuracy: 0.566
- F1 Score: 0.594
- Recall: 0.472

Despite improvements, performance remains far below in-domain results, highlighting a strong distribution gap.

4.3 Per-Task Analysis

Task	BERT F1	BioBERT F1
FCT	0.296	0.493
NOTA	0.348	0.651
FAKE	0.000	0.000

Table 2: Per-task performance on Med-HALT

BioBERT improves performance on reasoning tasks but both models fail on FAKE questions.

5 Discussion

The results highlight several key findings. First, both models perform strongly in-domain, indicating that hallucination detection is effective when training and

test distributions align. BioBERT provides consistent improvements, suggesting that domain-specific pretraining enhances semantic understanding.
Second, performance drops significantly on Med-HALT [2], demonstrating limited generalization. BERT exhibits strong bias toward predicting non-hallucinated answers, resulting in high precision but low recall.
Third, BioBERT substantially improves recall and overall F1, indicating better transfer to unseen tasks. However, the improvement is insufficient to fully bridge the distribution gap.
Finally, both models fail completely on FAKE tasks, revealing an inability to detect nonsensical or adversarial inputs. This aligns with findings from prior work [2, 5].

6 Conclusion

This project demonstrates that transformer-based classifiers can effectively detect hallucinations in medical QA within a controlled setting. However, generalization to more challenging scenarios remains limited.
BioBERT improves both in-domain and out-of-domain performance, confirming the importance of domain-specific knowledge. Nevertheless, the model still fails on adversarial and nonsensical inputs, indicating that hallucination detection requires deeper reasoning capabilities.
Future work could explore multi-task learning across MedHallu and Med-HALT, incorporate uncertainty estimation, or leverage generative self-checking methods such as SelfCheckGPT [5].

References

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